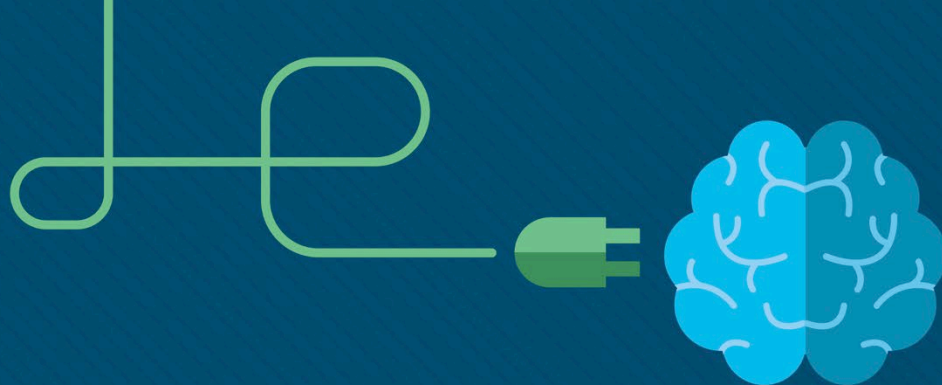




Module 6: EtherChannel

Switching, Routing and Wireless
Essentials v7.0 (SRWE)





Module Objectives

Module Title: EtherChannel

Module Objective: Describe configure and Troubleshoot EtherChannel on switched links.

Topic Title	Topic Objective
EtherChannel Operation	Describe EtherChannel technology.
Configure EtherChannel	Configure EtherChannel.
Verify and Troubleshoot EtherChannel	Troubleshoot EtherChannel.

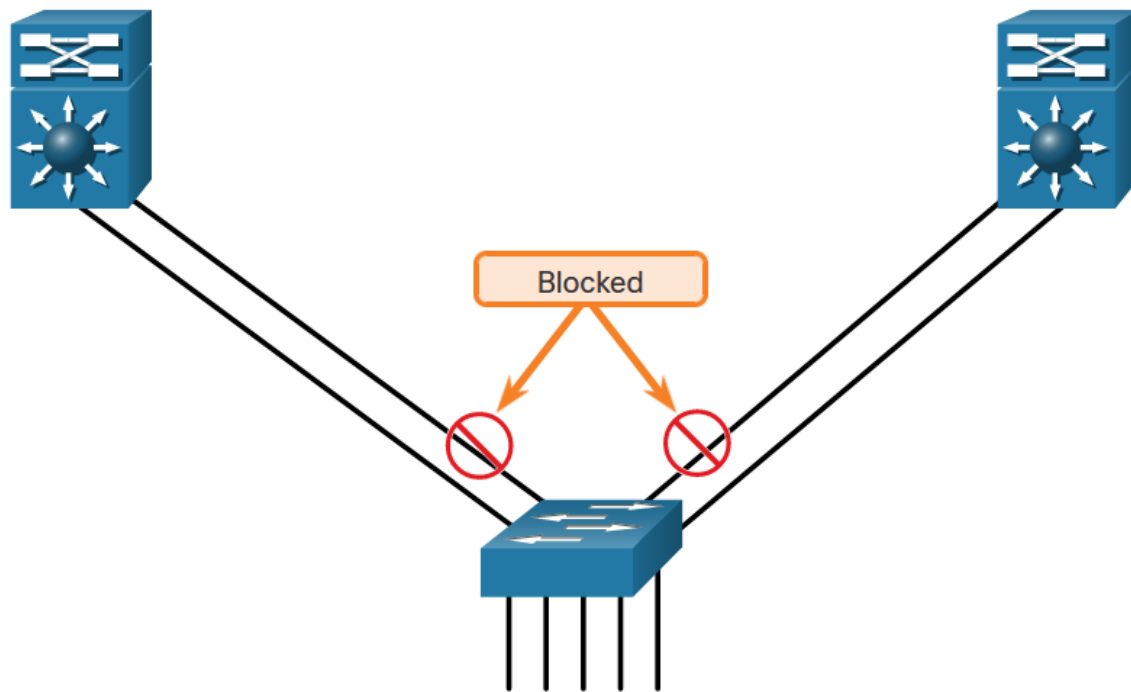


6.1 EtherChannel Operation

EtherChannel Operation

Link Aggregation

- bandwidth or redundancy
- Multiple links
- STP blocks redundant links to prevent switching loops
- link aggregation technology
- groups multiple physical Ethernet links together in one single logical link
- fault-tolerance, load sharing, increased bandwidth, and redundancy between switches, routers, and servers.

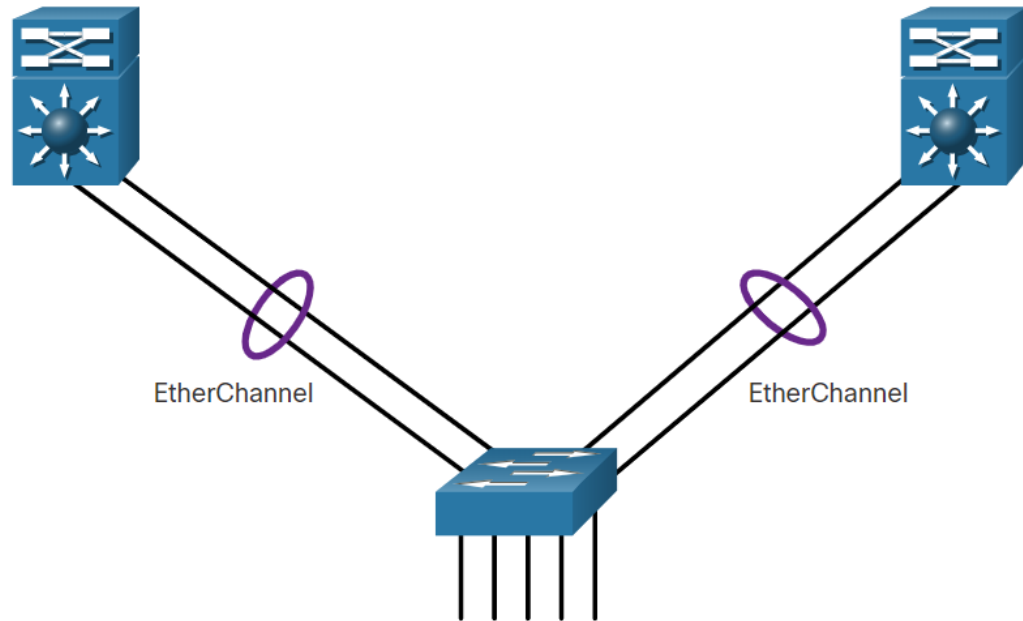


EtherChannel Operation

EtherChannel

EtherChannel technology was originally developed by Cisco as a LAN switch-to-switch technique of grouping several Fast Ethernet or Gigabit Ethernet ports into one logical channel.

When an EtherChannel is configured, the resulting virtual interface is called a port channel. The physical interfaces are bundled together into a port channel interface, as shown in the figure.



Advantages of EtherChannel

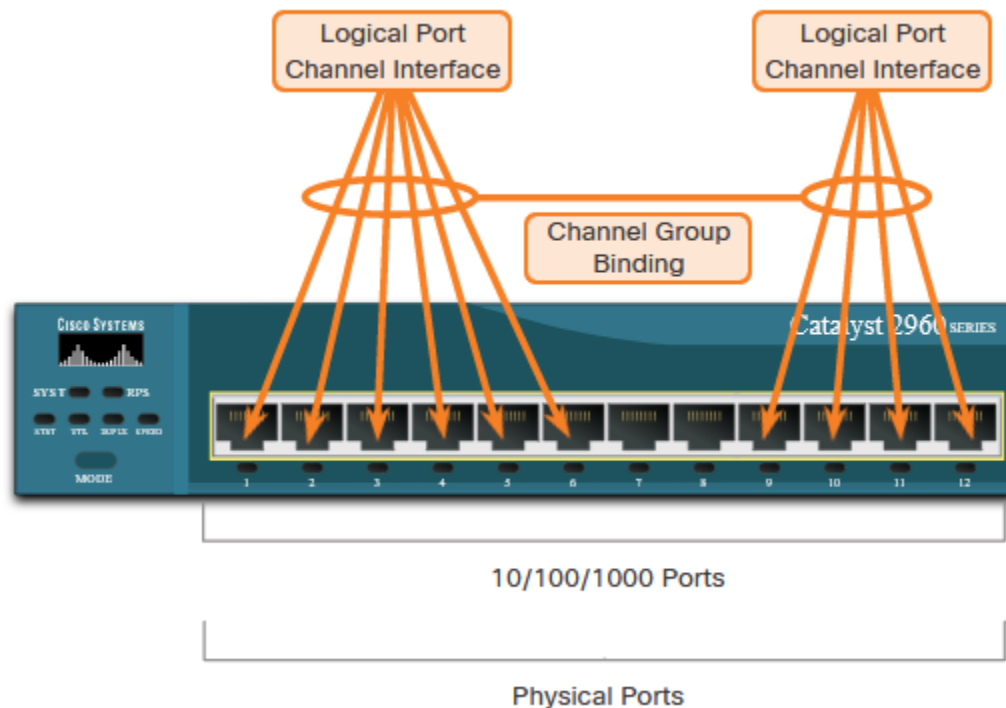
- Consistency - configuration on the Portchannel
- Cost no need for new devices or modules
- Load balancing takes place between links that are part of the same EtherChannel
- EtherChannel creates an aggregation that is seen as one logical link
- EtherChannel provides redundancy

EtherChannel Operation

Implementation Restrictions

EtherChannel has certain implementation restrictions, including the following:

- Interface types cannot be mixed.
- Consist of up to eight ports or less
- EtherChannel group member port configuration must be consistent
- has a logical port channel interface



EtherChannel Operation

AutoNegotiation Protocols

EtherChannels can be formed through negotiation using one of two protocols,

- Port Aggregation Protocol (PAgP)
- Cisco Proprietary Standard

- Link Aggregation Control Protocol (LACP).
- Standard IEEE 802.3ad

These protocols allow ports with similar characteristics to form a channel through dynamic negotiation with adjoining switches.

Note: It is also possible to configure a static or unconditional EtherChannel without PAgP or LACP.

- PAgP (pronounced “Pag - P”) is a Cisco-proprietary protocol that aids in the automatic creation of EtherChannel links.
- PAgP packets are sent between EtherChannel-capable ports to negotiate the forming of a channel
- When PAgP identifies matched Ethernet links, it groups the links into an EtherChannel.
- The EtherChannel is then added to the spanning tree as a single port.
- PAgP packets are sent every 30 seconds
and manages link additions and failures between two switches
- All ports must have the same speed, duplex setting, and VLAN information

EtherChannel Operation

PAgP Operation (Cont.)

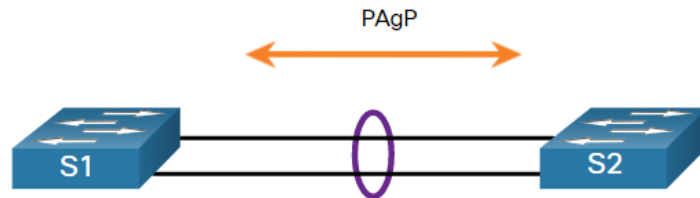
PAgP helps create the EtherChannel link by detecting the configuration of each side and ensuring that links are compatible so that the EtherChannel link can be enabled when needed.

The modes for PAgP as follows:

- **On** - This mode forces the interface to channel without PAgP. Interfaces configured in the on mode do not exchange PAgP packets.
- **PAgP desirable** - This PAgP mode places an interface in an active negotiating state in which the interface initiates negotiations with other interfaces by sending PAgP packets.
- **PAgP auto** - This PAgP mode places an interface in a passive negotiating state in which the interface responds to the PAgP packets that it receives but does not initiate PAgP negotiation.

EtherChannel Operation

PAgP Mode Settings Example



The table shows the various combination of PAgP modes on S1 and S2 and the resulting channel establishment outcome.

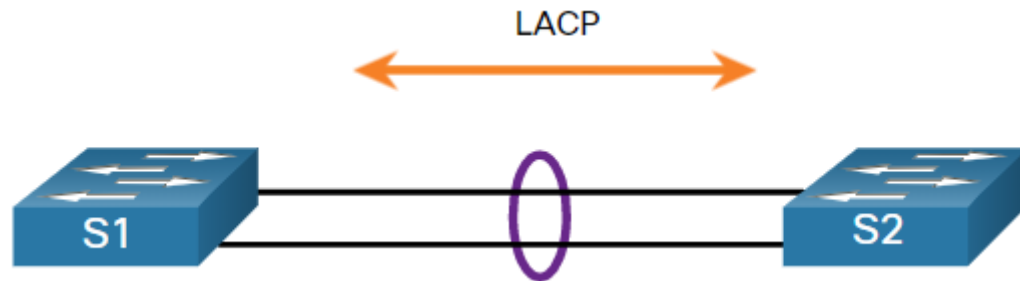
S1	S2	Channel Establishment
On	On	Yes
On	Desirable/Auto	No
Desirable	Desirable	Yes
Desirable	Auto	Yes
Auto	Desirable	Yes
Auto	Auto	No

LACP provides the same negotiation benefits as PAgP. LACP helps create the EtherChannel link by detecting the configuration of each side and making sure that they are compatible so that the EtherChannel link can be enabled when needed. The modes for LACP are as follows:

- **On** - This mode forces the interface to channel without LACP. Interfaces configured in the on mode do not exchange LACP packets.
- **LACP active** - This LACP mode places a port in an active negotiating state. In this state, the port initiates negotiations with other ports by sending LACP packets.
- **LACP passive** - This LACP mode places a port in a passive negotiating state. In this state, the port responds to the LACP packets that it receives but does not initiate LACP packet negotiation.



LACP Mode Settings Example



The table shows the various combination of LACP modes on S1 and S2 and the resulting channel establishment outcome.

S1	S2	Channel Establishment
On	On	Yes
On	Active/Passive	No
Active	Active	Yes
Active	Passive	Yes
Passive	Active	Yes
Passive	Passive	No